Euclidean Geometry In Mathematical Olympiads

P1.49(USAMO 2010/P1)

USAMO Chapter 1

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1031 posts

Mar 7, 2018, 6:23 PM • 2 #1

I'm having a lot of trouble with this one, and the hints don't really make too much sense. First, I don't see how to use a Simson line(or which one to use), and I don't understand what the angle hint means.

Provided Hint

Simson Lines from Y might help(not necessary though). For the other solution, begin by noting that the desired angle is $\angle PQY + \angle SRY - \angle QYR$.

Attachments:

Problem 1.49 (USAMO 2010/1). Let $AXYZB$ be a convex pentagon inscribed in a semicircle of diameter AB. Denote by P, Q, R, S the feet of the perpendiculars from Y onto lines AX, BX, AZ, BZ, respectively. Prove that the acute angle formed by lines PQ and RS is half the size of $\angle XOZ$, where O is the midpoint of segment AB. **Hint:** 661

QWERTYphysics
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Mar 8, 2018, 4:20 AM • 4 #2

The figure comes out a bit clogged.(it was for me)
Use Geogebra.
And try a neat angle chase.(Hopefully use Simson line)

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Mar 8, 2018, 12:05 PM • 3 #3

Oh okay, so dropping an altitude from Y to AB gives 2 simson lines. It tells you that the intersection of PQ and RS is at the diameter.

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Mar 8, 2018, 10:17 PM • 2 #4

Right,now try to angle chase in the most brutal way!!
It took me a bit of time to see it but I hope u see a lot of perpendiculars!!

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Mar 8, 2018, 10:30 PM • 3 #5

Right, the semicircle and diameter gives lots of right angles(and similar triangles).

Drunken_Master
327 posts

Mar 9, 2018, 4:17 AM • 3 #6

Solution

Let T be feet of Y on AB, $\angle XAZ = \alpha$ and $BX \cap AZ = M$.
Clearly, PQT and RST are $Y-$simson lines of $\triangle XAB$ and $\triangle ZAB$.
We have to show that $\angle QTR = \alpha$.
Firstly, $\angle QYR = \angle XMA = 90^\circ - \alpha$.
We need $\angle QTR = \alpha$, so it is required to prove that $\angle TQB + \angle TRA = 90^\circ$.
Indeed,

$$\angle TQB + \angle TRA = \angle PQX + \angle SRZ = 90^\circ - \angle PXY + 90^\circ - \angle SZY = 180^\circ - \angle YBA - \angle YAB = 90^\circ$$

Hence, we are done. $\square$

This post has been edited 1 time. Last edited by WizardMath, Mar 9, 2018, 6:50 AM
Reason: Hidden solution.

277546
1607 posts

Apr 11, 2018, 8:24 PM • 3 *#8*

Solution

This post has been edited 2 times. Last edited by 277546, Feb 25, 2020, 4:20 PM

Vrangr
1600 posts

Apr 13, 2018, 3:44 AM • 5 *#9*

Solution

> *realquarterb wrote:*
> P.S. That diagram took forever to draw

It was alright.

@below, that was my intent

This post has been edited 1 time. Last edited by Vrangr, Sep 12, 2018, 3:43 AM

DynamoBlaze
170 posts

Sep 11, 2018, 2:29 PM • 4 *#10*

> *Vrangr wrote:*